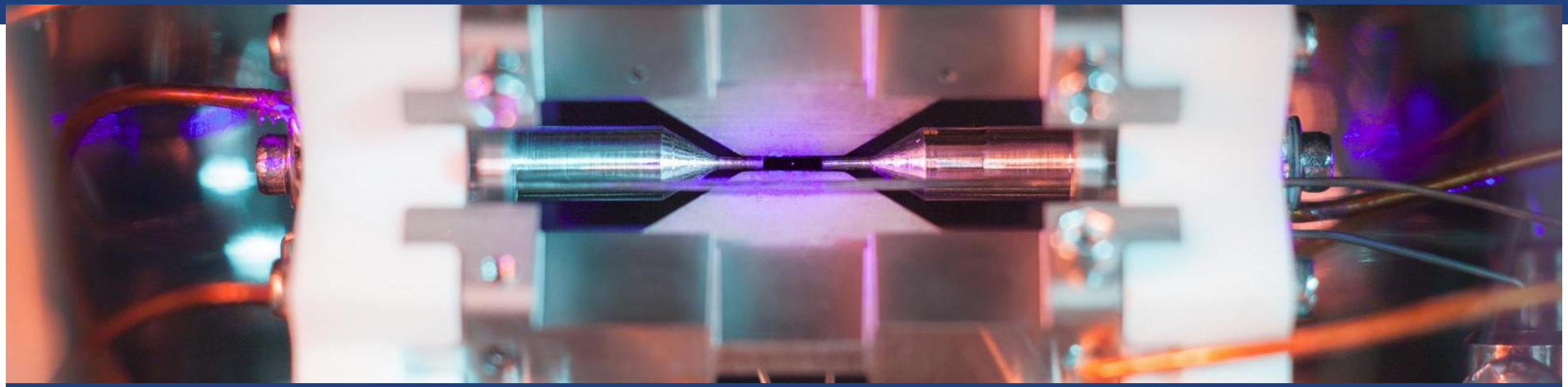
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Quantum-enhanced sensing of a single-ion mechanical oscillator

Tommaso Bonfanti

Quantum Sensing and Metrology FS26 – 03.06.2026

LETTER

https://doi.org/10.1038/s41586-019-1421-y

Quantum-enhanced sensing of a single-ion mechanical oscillator

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Agenda

1. Introduction and theory
2. State preparation
3. Quantum-enhanced sensing
4. Frequency tracking
5. Conclusions and outlook







Agenda

1. Introduction and theory



2. State preparation



3. Quantum-enhanced sensing



4. Frequency tracking



5. Conclusions and outlook



Motivation

- Harmonic oscillator systems are fundamental building blocks in quantum technology
- Achieving better precision is critical for clocks, gyroscopes, gravitational wave detection and QIP
- Frequency drifts in trapped ion systems restrict gate fidelity and scalability

Quantum – enhanced sensing

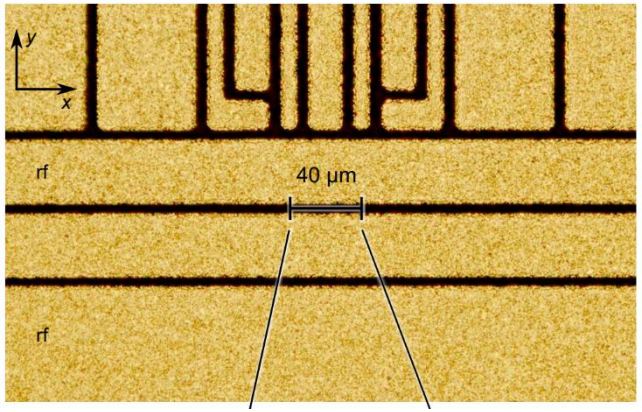


Fig. 1: Microscope image of the linear surface-electrode ion trap design on which the experiment is based (Extracted from Brown et al., Nature 471, 196–199 (2011))

Number state interferometry

Standard Quantum Limit

$$\delta\phi \sim \frac{1}{\sqrt{n}}$$

- Harmonic oscillator with frequency $\omega + \delta\omega(t)$
- To achieve quantum enhancement, use a non-classical superposition state $|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |n\rangle)$

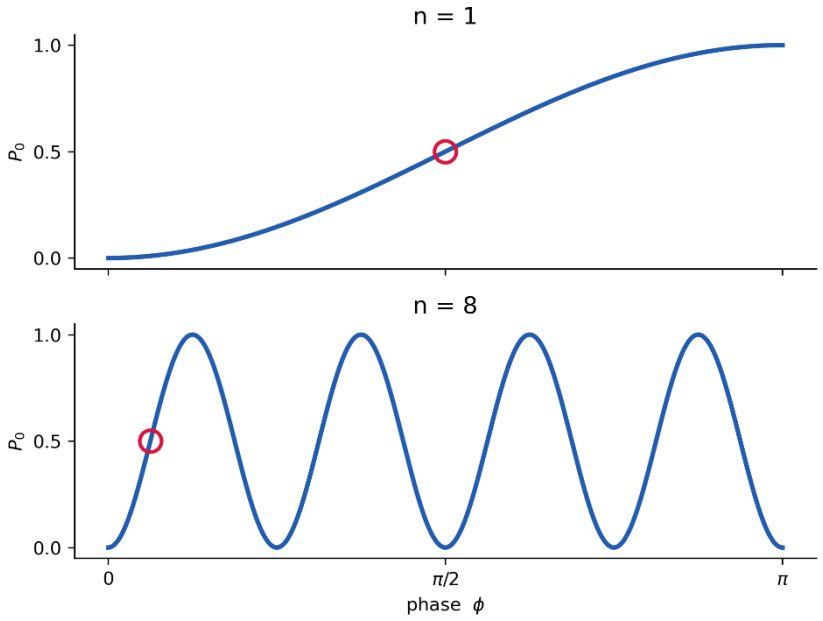
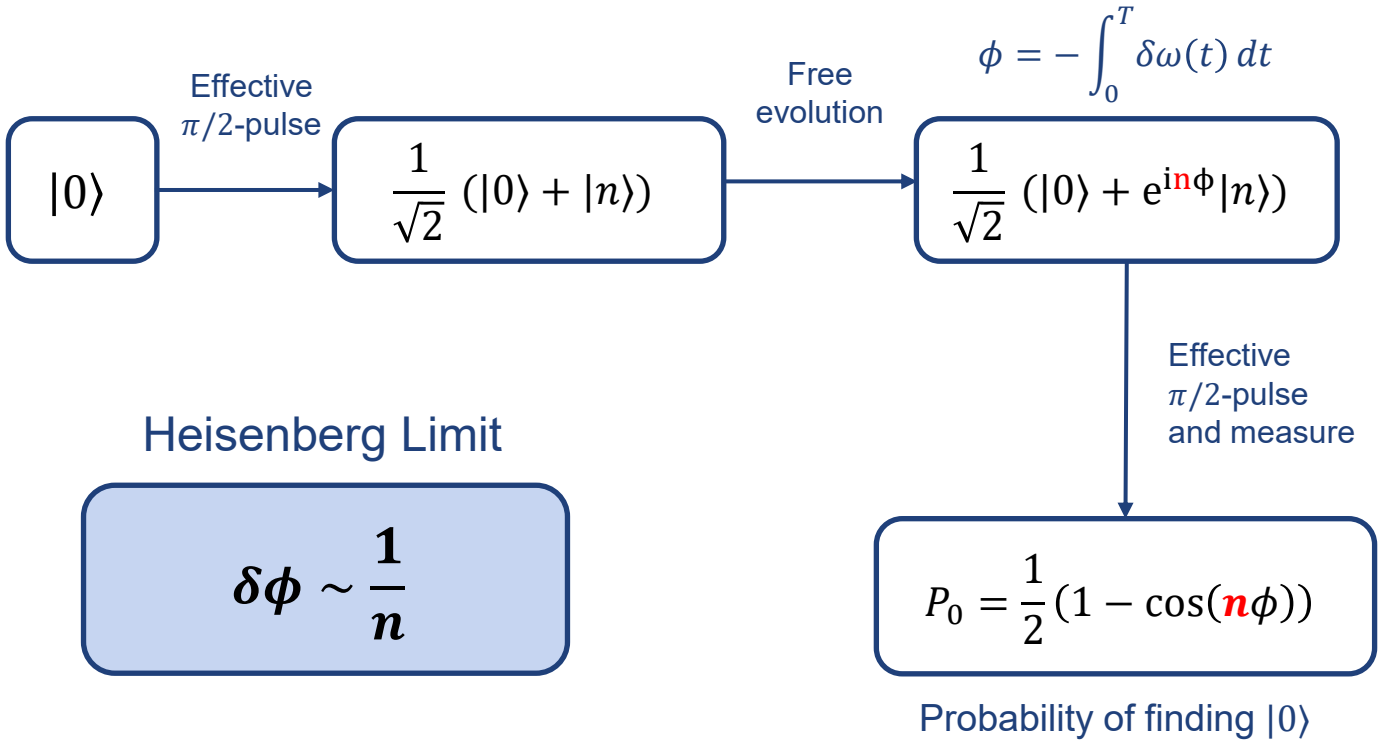


Fig. 2: Population fringe $P_0 = \frac{1}{2}(1 - \cos(n\phi))$ for n = 1 and n = 8. The slope at the steepest point scales as n/2, giving smaller phase uncertainty $\delta\phi \sim 1/n$.



Heisenberg Limit

$$\delta\phi \sim \frac{1}{n}$$

Why is this the best state?

Fixed resources: consider states with quantum number up to n

Quantum Cramer-Rao Bound

Which state minimizes phase uncertainty $\delta\phi$?

$$(\delta\phi)^2 \geq \frac{1}{F_Q(\phi)_{|\Psi\rangle}}$$

$$F_Q(\phi)_{|\Psi\rangle} = 4\left(\Delta\hat{G}\right)^2_{|\Psi\rangle}$$

Free evolution: the generator is $\hat{G} = \hat{n}$

Quantum speed limit

Which state reaches an orthogonal state in the shortest time?

$$\tau_\perp \geq \frac{\pi\hbar}{2\Delta E_{|\Psi\rangle}}$$

Harmonic oscillator: energy given by $\hat{E} = \hbar\omega\hat{n}$

The widest spread Δn is achieved by $|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |n\rangle) \Rightarrow \Delta n = n/2$

$$\Longrightarrow F_Q(\phi)_{|\Psi\rangle} = n^2 \text{ and } \tau_\perp = \frac{\pi}{n\omega}$$

Intuition for Heisenberg scaling

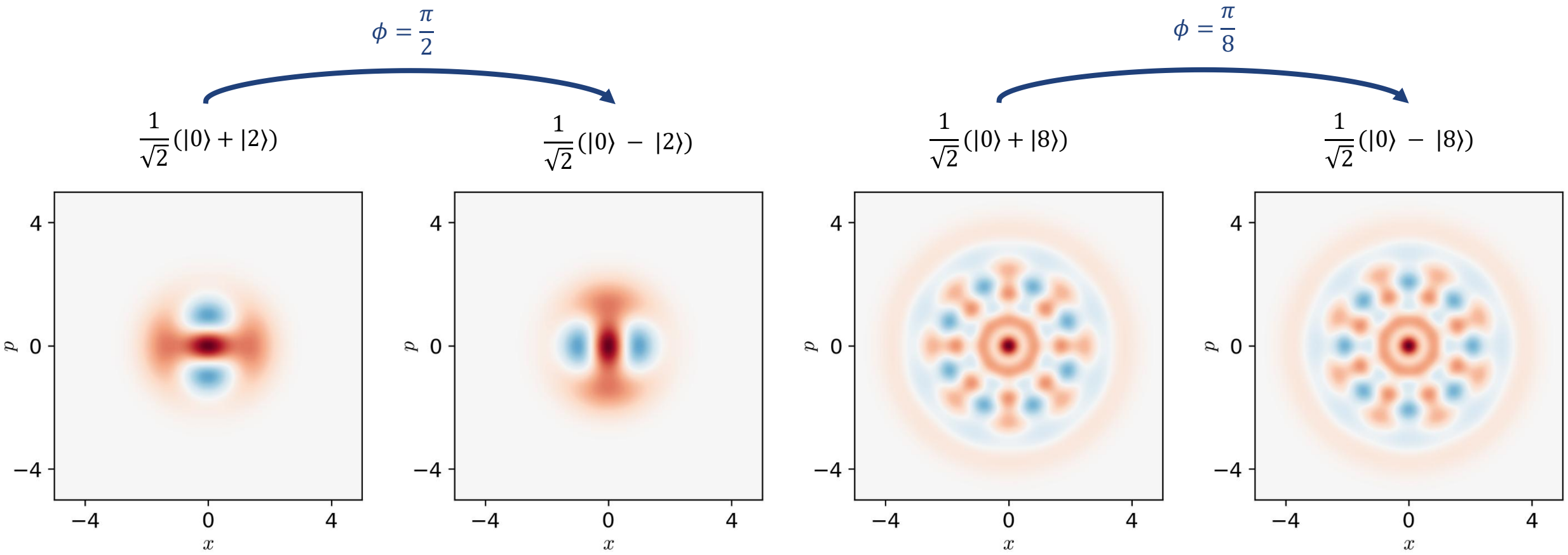


Fig. 3: Wigner functions of the superposition states $(|0\rangle + e^{in\phi}|n\rangle)/\sqrt{2}$ for n = 2 and n = 8. In each pair, the left panel shows the state at φ = 0 and the right panel the same state after evolving by ϕ = π/n, where it becomes orthogonal to the initial state.

Goals of the paper

- Preparing **non-classical superposition states** $|\mathbf{0}\rangle + |\boldsymbol{n}\rangle$ of the axial mode of a single trapped ion
- Demonstrating the improved phase sensitivity of these states, achieving **Heisenberg scaling**
- Applying the **enhanced sensitivity** to tracking of the trap motional frequency

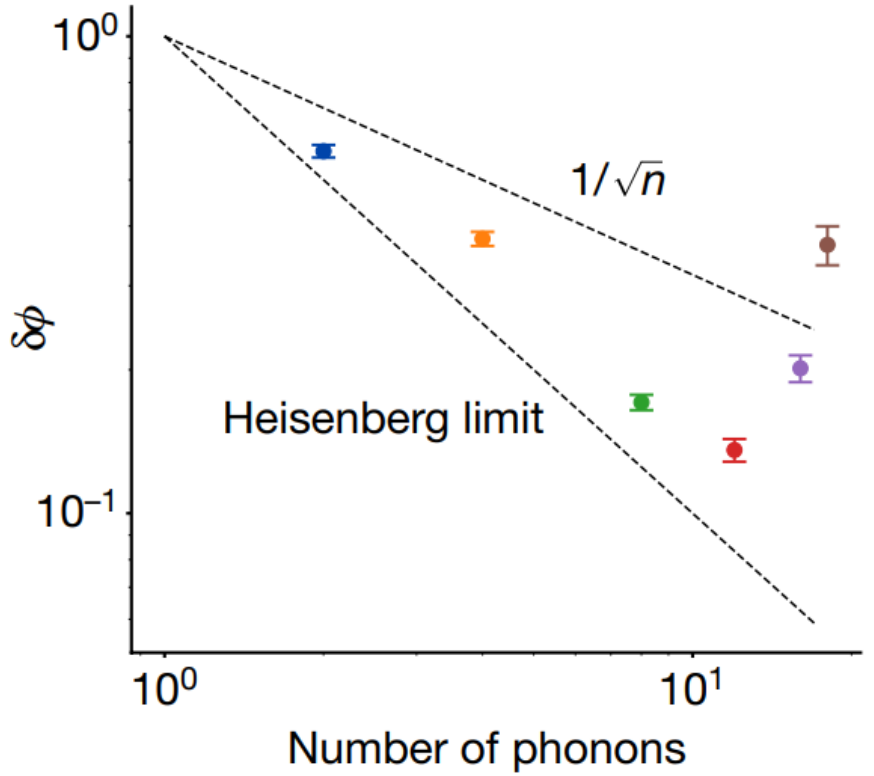


Fig. 4: Experimentally determined phase sensitivity $\delta\phi$ as a function of order n (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Agenda

1. Introduction and theory
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Experimental platform

- Cryogenic (4K) linear surface electrode trap holding a single $^{9}Be^{+}$ ion
  - Internal 3 level system within the $^{2}S_{1/2}$ ground state hyperfine manifold
  - Motional state $|n\rangle$ of the ion along the axial mode at $\omega \simeq 2\pi \times 7.2$ MHz
- Microwave carrier transition couples $|\uparrow\rangle|k\rangle \leftrightarrow |\mathrm{aux}\rangle|k\rangle$
- Sideband transitions exchange energy between the motional and internal states
- Readout of internal state via state-dependent fluorescence

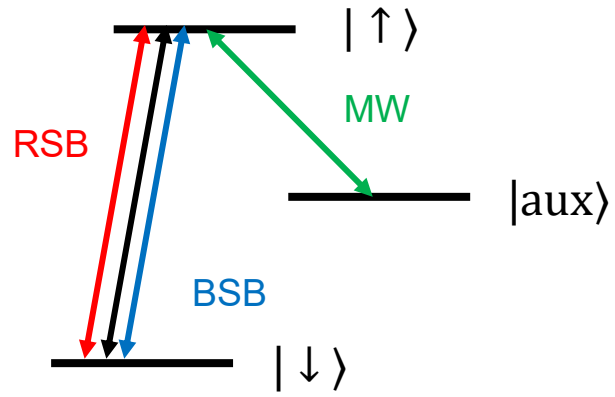


Fig. 4: sketch of the 3 level system in the electronic structure of the ion

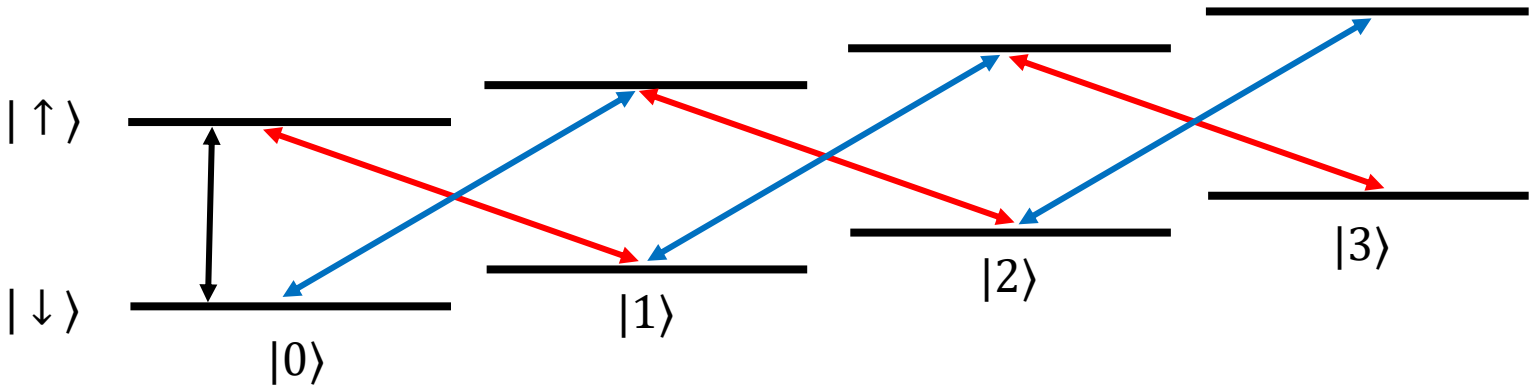


Fig. 5: sketch of the joint Hilbert space of the internal and motional degrees of freedom

State preparation - $|n\rangle$

- Doppler cooling + sideband cooling to prepare $|\downarrow\rangle|0\rangle$
- Carrier π-pulse to bring population to $|\uparrow\rangle|0\rangle$
- Each sideband π-pulse adds a quantum of motion

  - Achieved Fock states up to $|\mathrm{n} = 100\rangle$

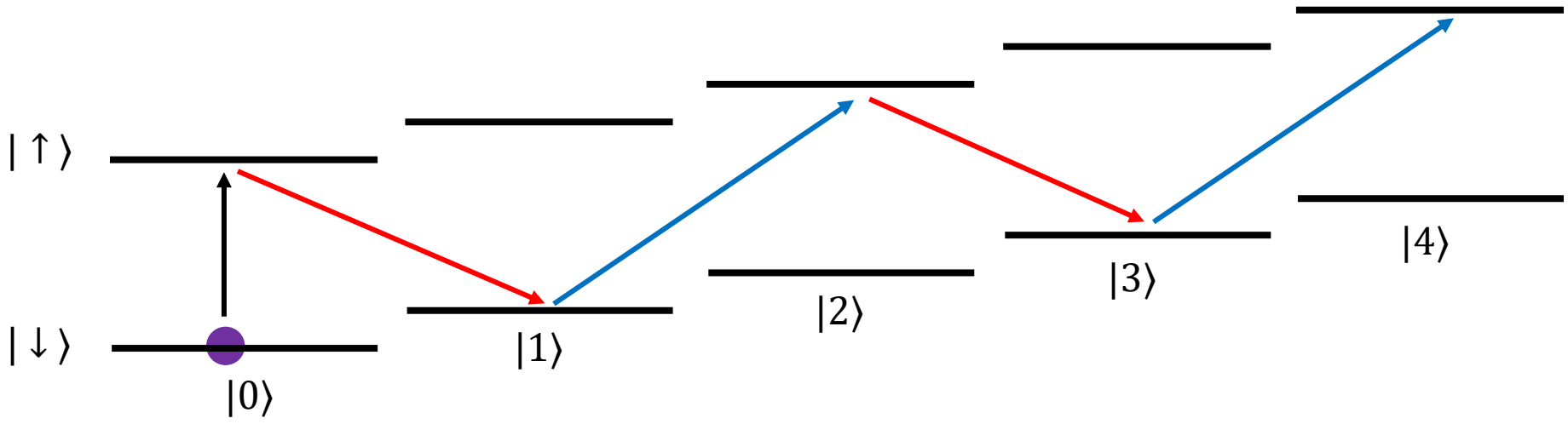
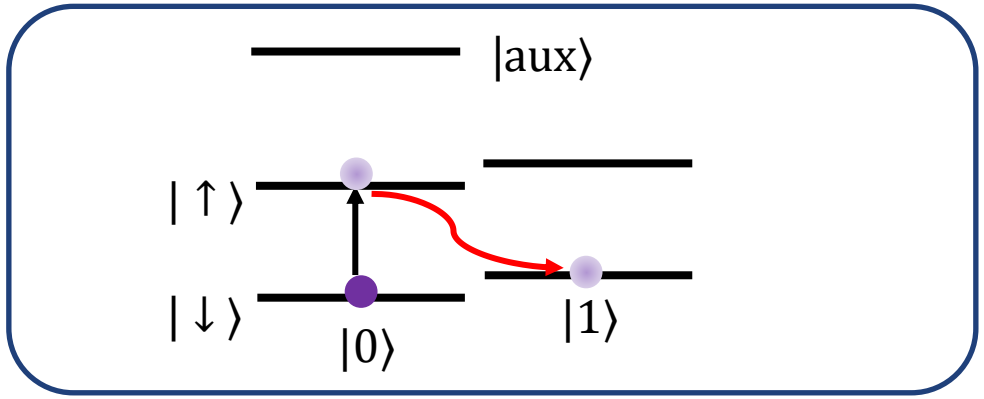


Fig. 6: sketch of the preparation of a Fock state $|n\rangle$

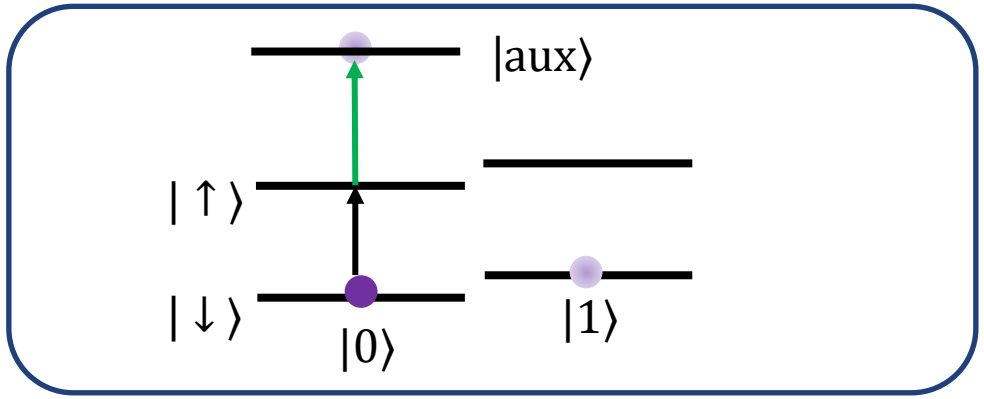
State preparation - $|0\rangle + |n\rangle$

1. RSB π/2-pulse creates superposition $|\downarrow\rangle|1\rangle + |\uparrow\rangle|0\rangle$



State preparation - $|0\rangle + |n\rangle$

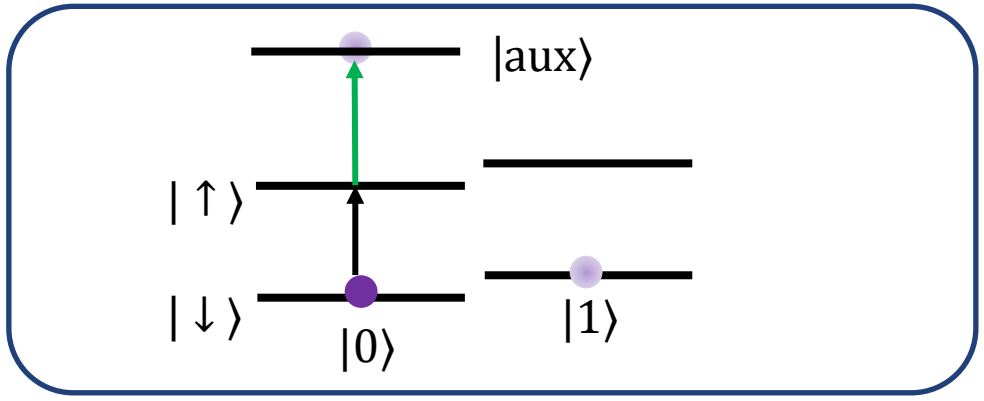
1. RSB π/2-pulse creates superposition $|\downarrow\rangle|1\rangle + |\uparrow\rangle|0\rangle$
2. Microwave π-pulse to shelve the population in $|\text{aux}\rangle|0\rangle$

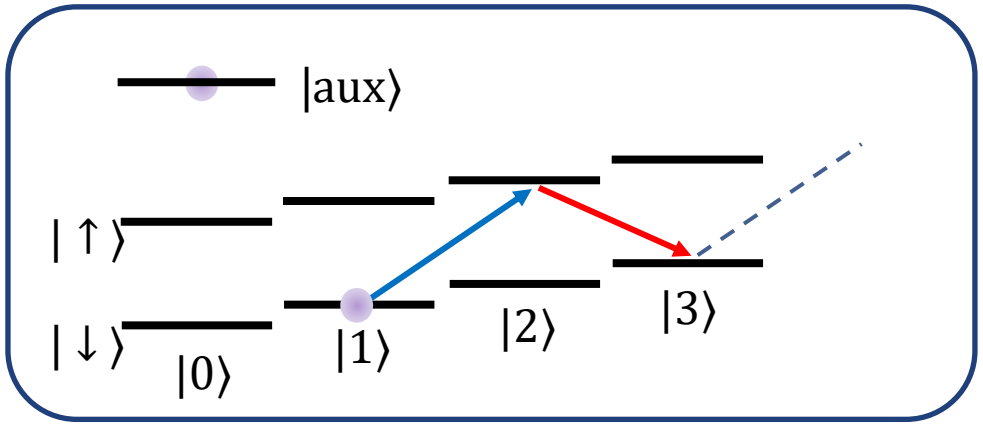


State preparation - $|0\rangle + |n\rangle$

1. RSB π/2-pulse creates superposition $|\downarrow\rangle|1\rangle + |\uparrow\rangle|0\rangle$
2. Microwave π-pulse to shelve the population in $|\text{aux}\rangle|0\rangle$
3. Climb the ladder until we reach, for the desired n, the state

$$|\downarrow\rangle|n-1\rangle + |\text{aux}\rangle|0\rangle$$



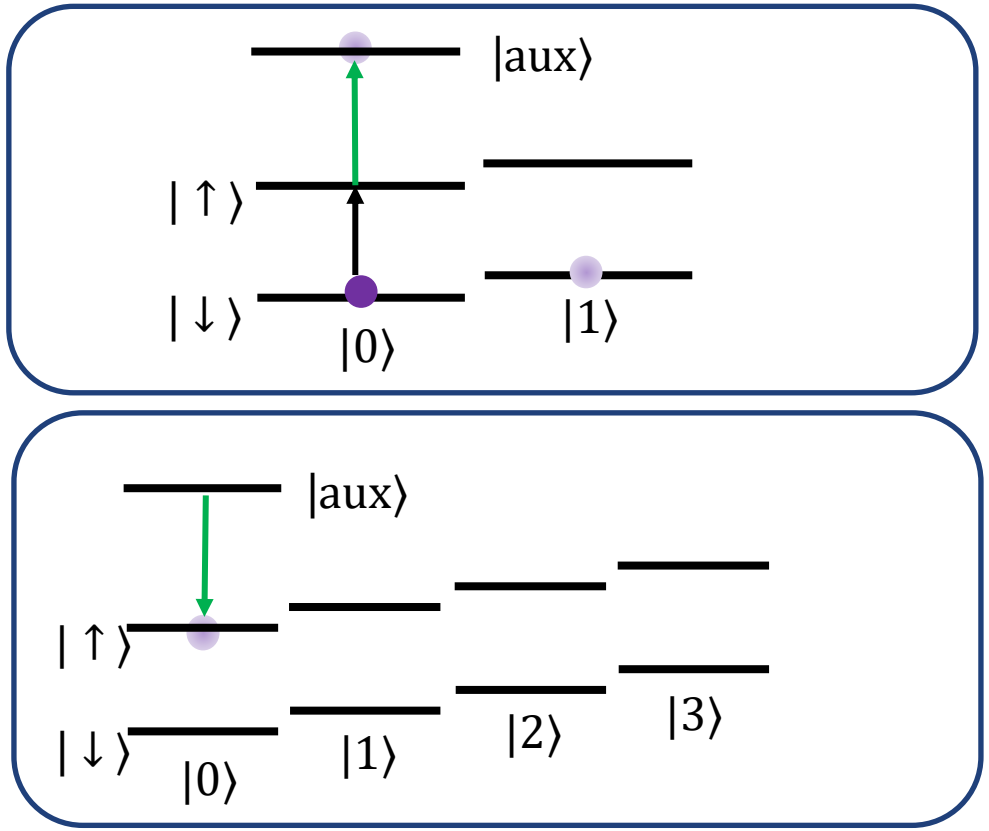


State preparation - $|0\rangle + |n\rangle$

1. RSB π/2-pulse creates superposition $|\downarrow\rangle|1\rangle + |\uparrow\rangle|0\rangle$
2. Microwave π-pulse to shelve the population in $|\text{aux}\rangle|0\rangle$
3. Climb the ladder until we reach, for the desired n, the state

$$|\downarrow\rangle|n-1\rangle + |\text{aux}\rangle|0\rangle$$

4. Microwave π-pulse to bring back the population in $|\uparrow\rangle|0\rangle$



State preparation - $|0\rangle + |n\rangle$

1. RSB π/2-pulse creates superposition $|\downarrow\rangle|1\rangle + |\uparrow\rangle|0\rangle$
2. Microwave π-pulse to shelve the population in $|aux\rangle|0\rangle$
3. Climb the ladder until we reach, for the desired n, the state

$$|\downarrow\rangle|n-1\rangle + |aux\rangle|0\rangle$$

4. Microwave π-pulse to bring back the population in $|\uparrow\rangle|0\rangle$
5. BSB π-pulse creates the final state:

$$|\Psi\rangle = |\uparrow\rangle(|0\rangle + |n\rangle)$$

- Achieved superpositions up to $n = 18$

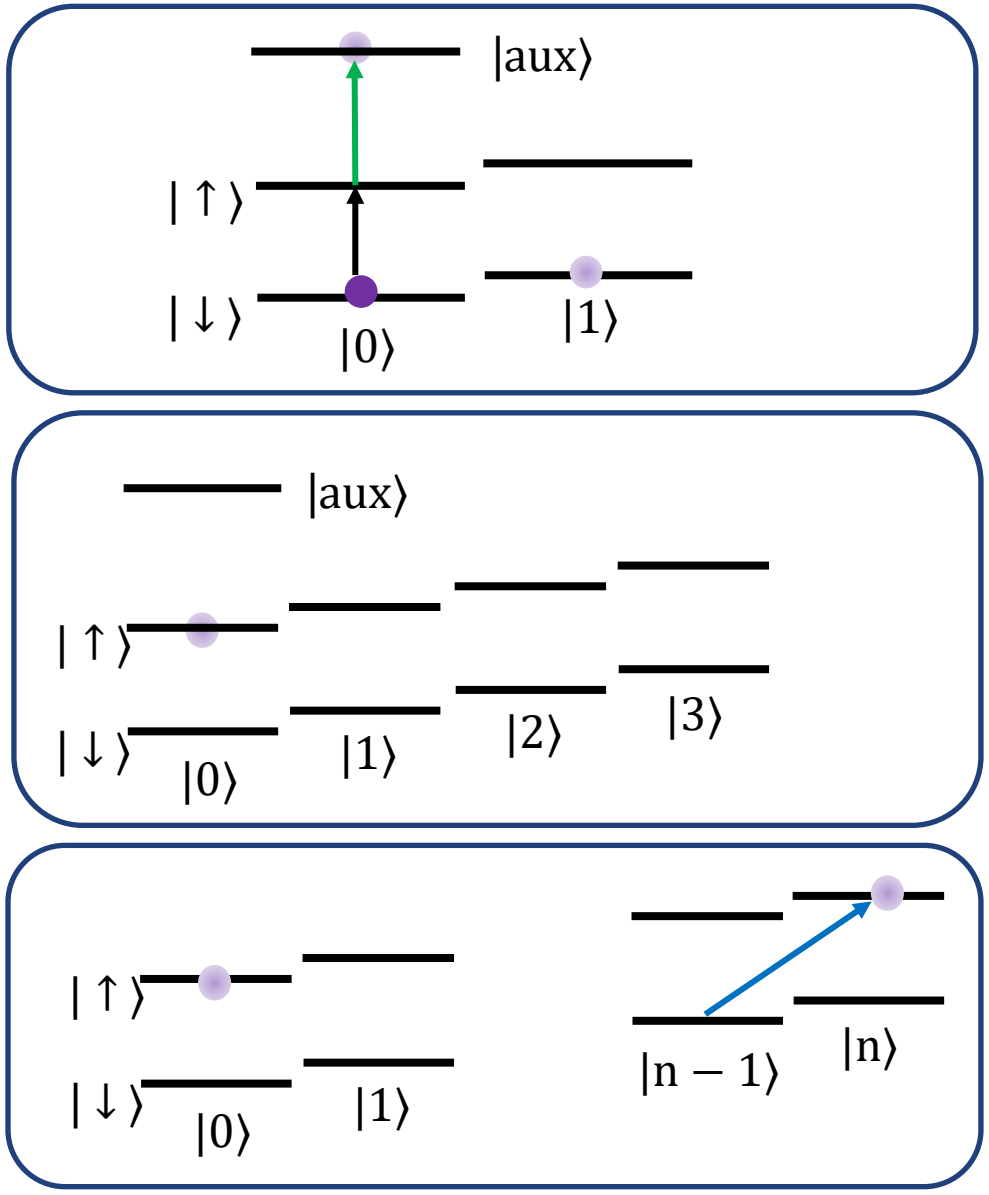


Fig 7: sketch of the state preparation process for a superposition state used in the paper

Agenda

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Number state interferometer - implementation

- Characterization by inducing known trap frequency changes $\pm\Delta\omega$

$$\phi = -2\Delta\omega T$$

Preparing the superposition state $\frac{1}{\sqrt{2}}(|0\rangle + |n\rangle)$

Echo pulse: flipping the sign of the accumulated phase. Common drifts canceled by symmetry; ±Δω add constructively.

Mapping accumulated phase into a population $P_{\downarrow} = \frac{1}{2}(1 - \cos(n\phi))$

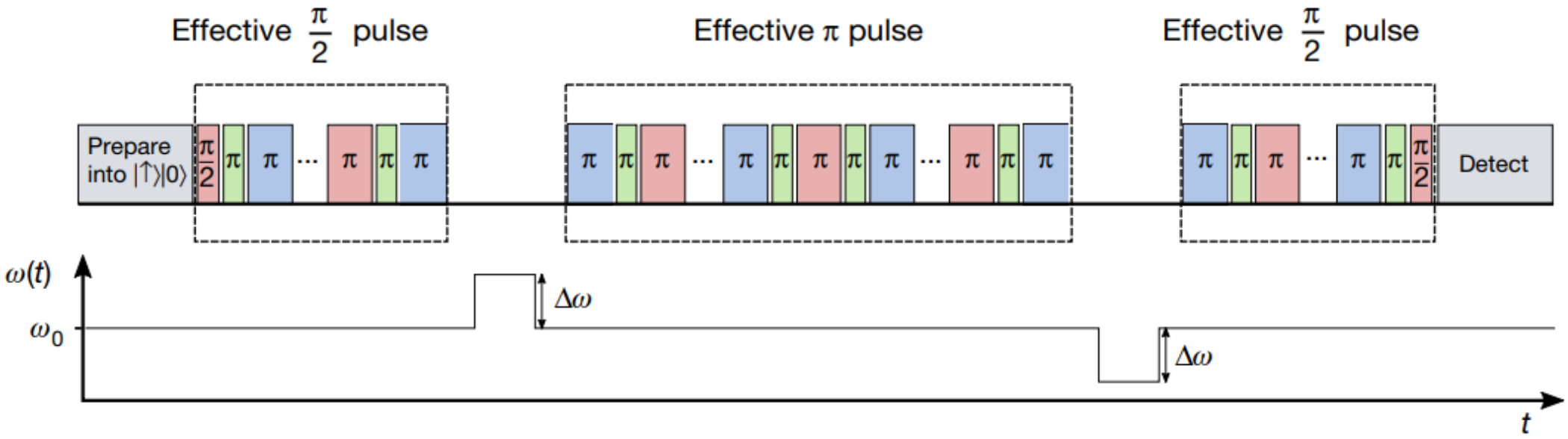


Fig 8: Pulse sequences and trap frequencies for number-state interferometers (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Results - Effective sensitivity

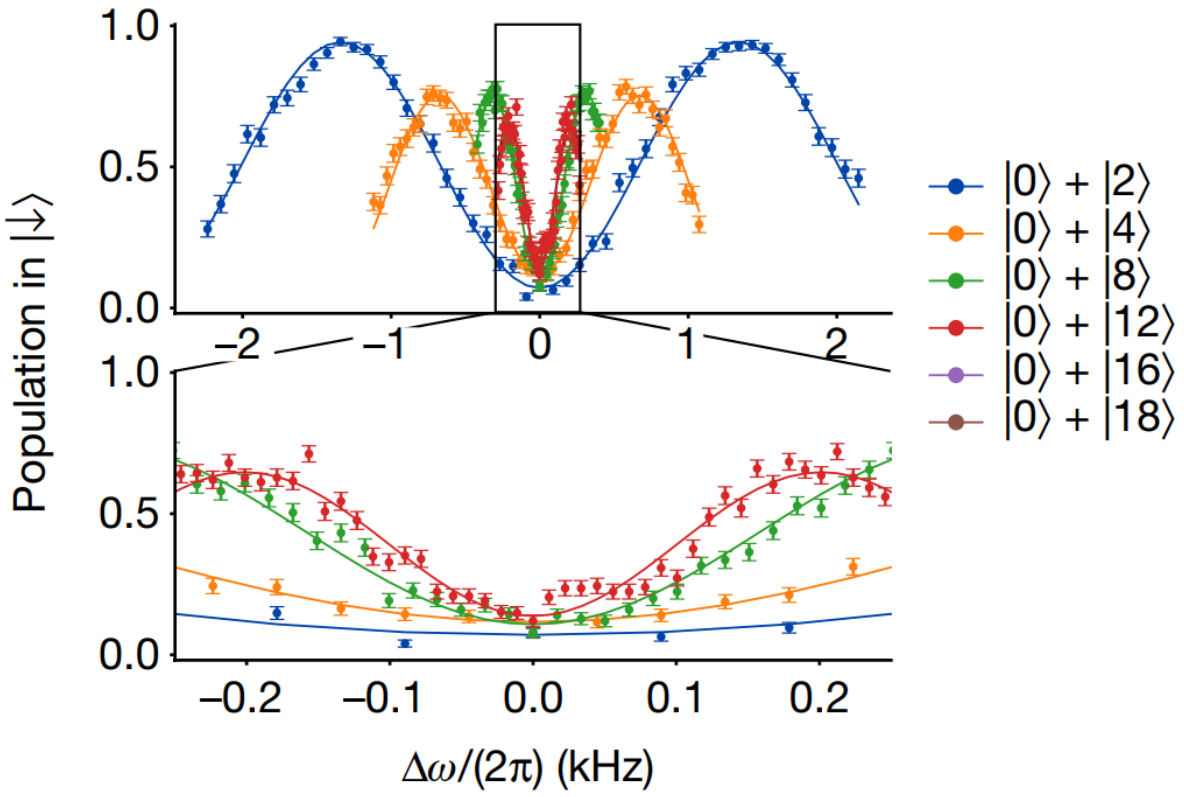


Fig 9: Interference fringes for number-state interferometers. Fringe spacing narrows as 1/n (Heisenberg scaling), while contrast decreases with larger n due to accumulated pulse errors and mode-frequency instability. (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Effective sensitivity

$$\delta\phi_{eff} \sim \frac{1}{nC(n)}$$

- C(n) is the **contrast** of the interference fringes, which drops for large n due to experimental imperfections
  - Accumulated pulse infidelity
  - Frequency instability over many shots
- The optimal point is found to be n=12
  - 6.4(4) dB sensitivity enhancement compared to an ideal classical interferometer with the same resources
  - 17(4) dB improvement compared to an ideal n=1 interferometer

Results - Heisenberg scaling

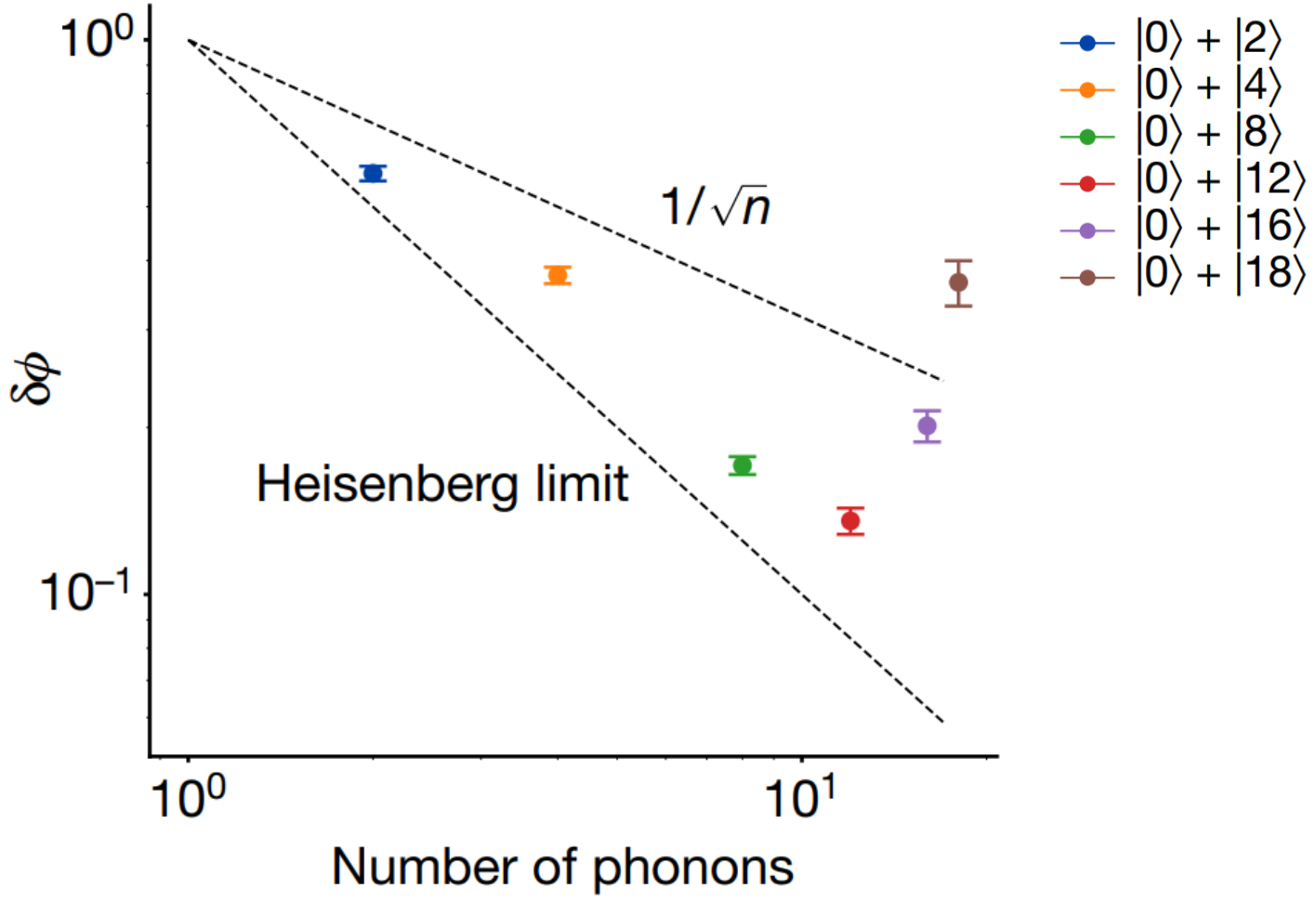


Fig 10: Experimentally determined phase sensitivity $\delta\phi$ as a function of order n (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Agenda

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Frequency tracking

- **Goal:** follow the trap frequency drifts over time
- The local oscillator frequency is our best current estimate of ω
- Perform 2 Ramsey experiments with opposite phases $\pm\pi/2$ in the final pulse
  - Signed feedback error signal: $\Delta P = P_{\downarrow}^{+} - P_{\downarrow}^{-} \propto n \cdot \delta\omega$
- Real implementation: auto-balanced Ramsey to cancel phase offsets caused by Stark shift and state preparation

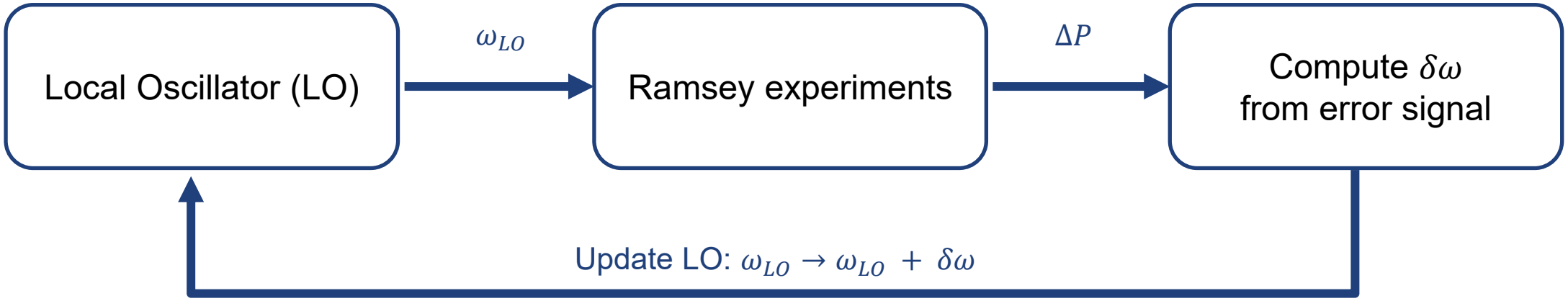


Fig 11: simplified sketch of the feedback loop used in the paper to track the ion motional frequency

Frequency tracking - results

- Interleaved experiments for n=2,4,6,8: identical drifting conditions

- **Quantum advantage:** higher n reaches a given precision faster
  - The $n = 8$ interferometer achieves its best precision of 2.6×10^{-6} (19 Hz at 7.2 MHz) after $\tau \simeq 23s$
  - Running only $n = 8$ achieves same precision in $\tau \simeq 4s$

- For long averaging times, frequency drifts dominate the uncertainty

Allan Deviation

Frequency precision vs averaging time

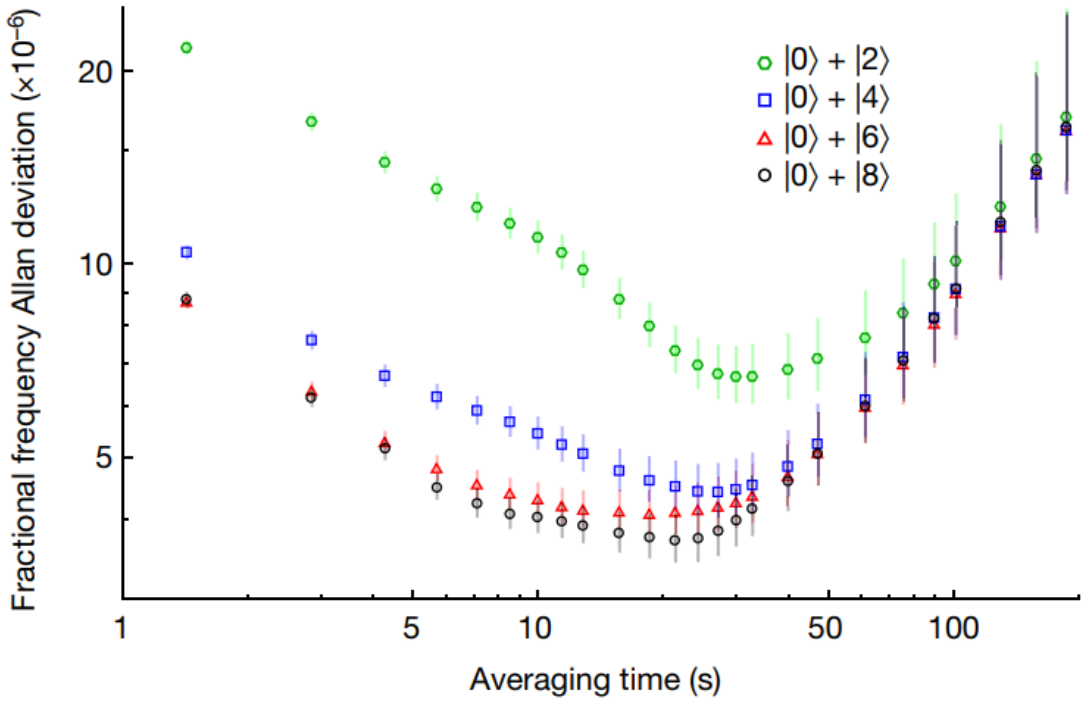


Fig 12: Allan standard deviation of tracked fractional trap frequencies versus averaging time for interleaved experiments with n = 2, n = 4, n = 6 and n = 8 interferometers . (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

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Conclusions

- Demonstrated high-fidelity **state preparation** of:
  - Fock states up to $|\mathrm{n} = 100\rangle$
  - Superposition states $|0\rangle + |n\rangle$ up to $n = 18$
- Characterized the **quantum-enhanced sensitivity** of these states, approaching Heisenberg scaling and achieving for $n = 12$:
  - 6.4(4) dB enhancement compared to an ideal classical interferometer
  - 17(4) dB enhancement to an ideal $|0\rangle + |1\rangle$ interferometer
- Tracked frequency drifts, achieving a minimum fractional uncertainty of 2.6×10^{-6} using an $n = 8$ quantum interferometer.

Outlook

- **Generalizing** number-state interferometry to achieve quantum-enhanced phase sensitivity in other harmonic oscillator systems

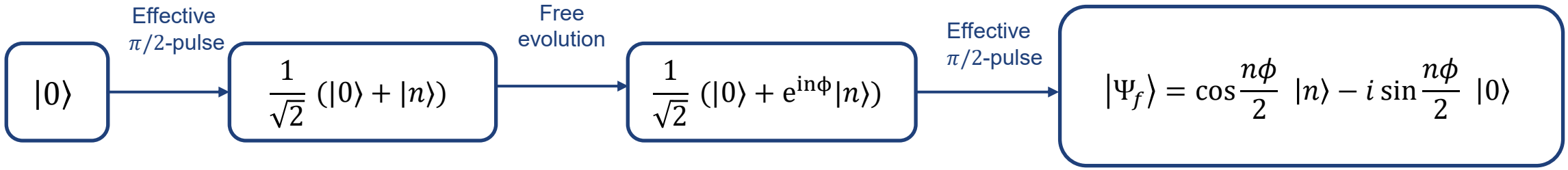
- Improving characterization of **motional decoherence** in trapped-ion processors, where it remains a key limiting factor
  - Spectator ions continuously monitoring motional frequency drifts for real-time recalibration

Ref. Majumder et al., *npj Quantum Information* **6**, 19 (2020)

THANK YOU!
QUESTIONS?

SUPPLEMENTARY SLIDES

Heisenberg scaling - derivation



Probability to measure the state $|0\rangle$ is given by: $P_0 = |\langle 0|\Psi_f\rangle|^2 = \frac{1}{2}(1-\cos n\phi)$

Best sensitivity is achieved at the steepest slope point, where $|\frac{\partial P_0}{\partial \phi}| = |\frac{n}{2}\sin(n\phi)| = \frac{n}{2}$

If we assume a perfect experimental setup, the measurement will be **Projection Noise Limited**: each shot is a Bernoulli trial with probability P_0 of measuring the state $|0\rangle$, resulting in standard deviation

$$\Delta P = \sqrt{P_0(1-P_0)} = \frac{1}{2}$$

Sensitivity is then given by

$$\delta\phi = \frac{\Delta P}{|\partial P_0/\partial\phi|} = \frac{1}{n}$$

Maximum variance state

Consider the state $|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |n\rangle)$

In the interferometer, the interaction is given by the generator $\widehat{\mathrm{G}} = \hat{n}$

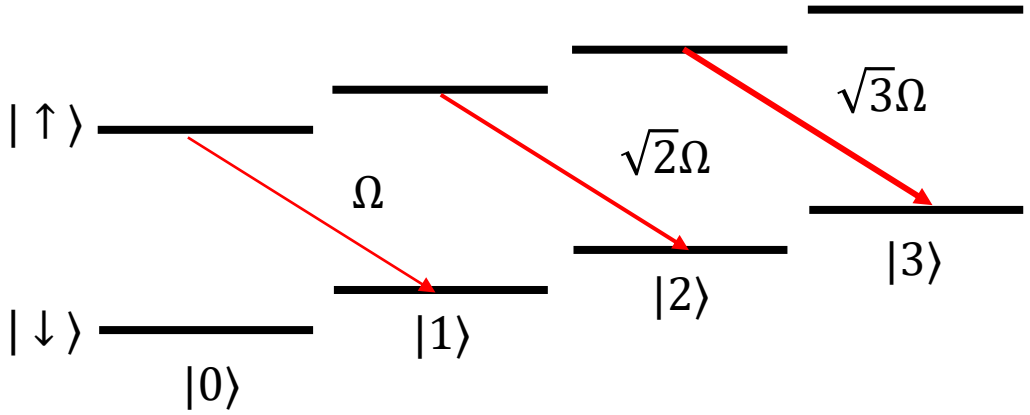
If we calculate the variance of the generator on this state, we get:

$$\langle \hat{n} \rangle = \frac{n}{2} \qquad \langle \hat{n}^2 \rangle = \frac{n^2}{2} \qquad \Rightarrow \qquad \mathrm{Var}(\hat{n}) = \frac{n^2}{2} - \left(\frac{n}{2}\right)^2 = \frac{n^2}{4}$$

$$\rightarrow \quad \boldsymbol{F_Q(\phi)_{|\Psi\rangle} = n^2} \quad \rightarrow \quad \boldsymbol{\delta\phi = \frac{1}{n}}$$

Sideband flopping of number states

- How do they know they actually prepared the right state $|n\rangle$?
  - Jaynes Cummings ladder gives n-dependent Rabi Frequency



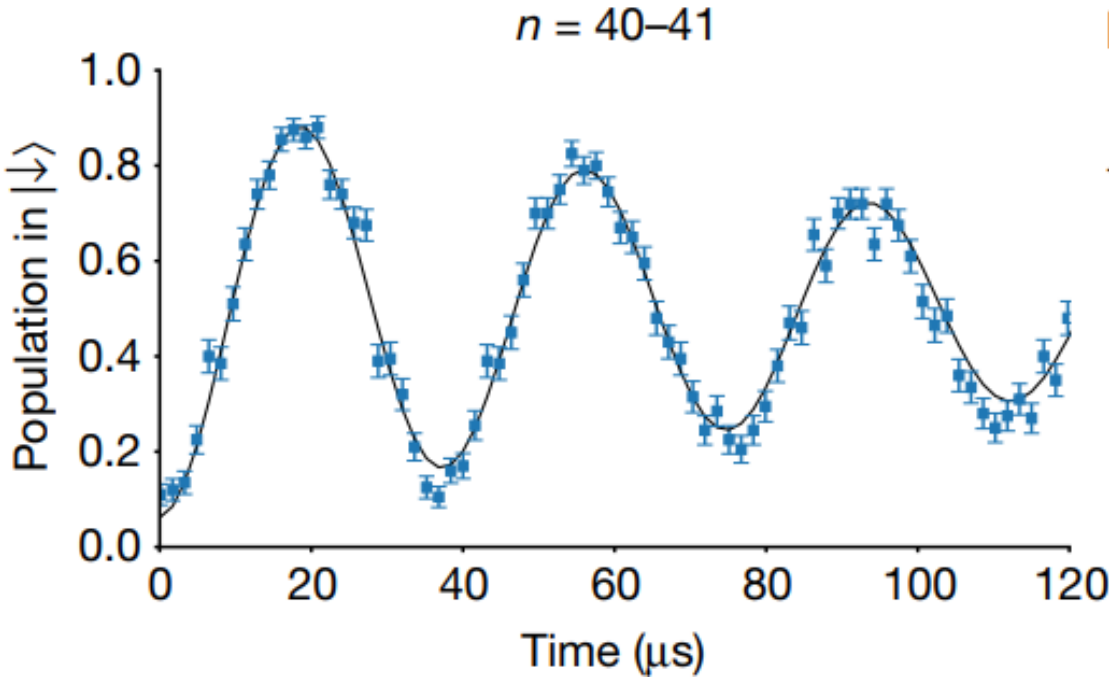


Fig S.1: Experimental observation of Red Sideband (RSB) flopping across various orders. First-order RSB transitioning from $|\uparrow, n = 40\rangle$ to $|\downarrow, n = 41\rangle$ (200 averages)..
(Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Sideband flopping on number states – $|n = 100\rangle$

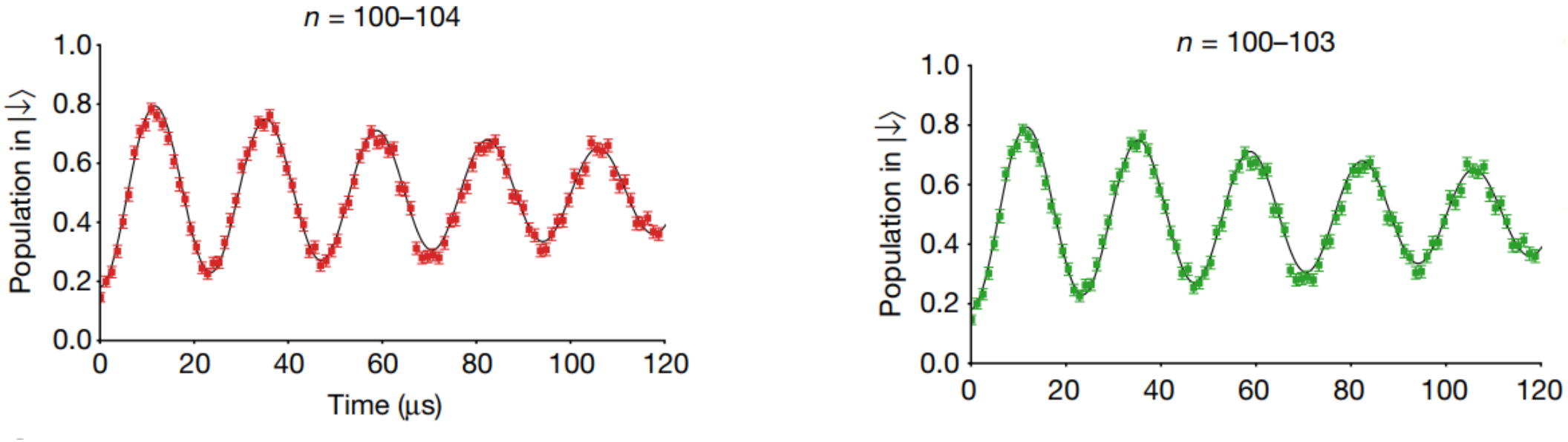


Fig S.2: Experimental observation of Red Sideband (RSB) flopping across various orders.
(b) Fourth-order RSB flopping from $|\uparrow, n = 100\rangle$ to $|\downarrow, n = 104\rangle$ (500 averages). (c) Third-order RSB flopping between $|\uparrow, n = 100\rangle$ and $|\downarrow, n = 103\rangle$. Solid lines represent theoretical fits used to determine the Rabi frequency Ω, initial contrast, and exponential decay constant γ.
(Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Higher order sidebands

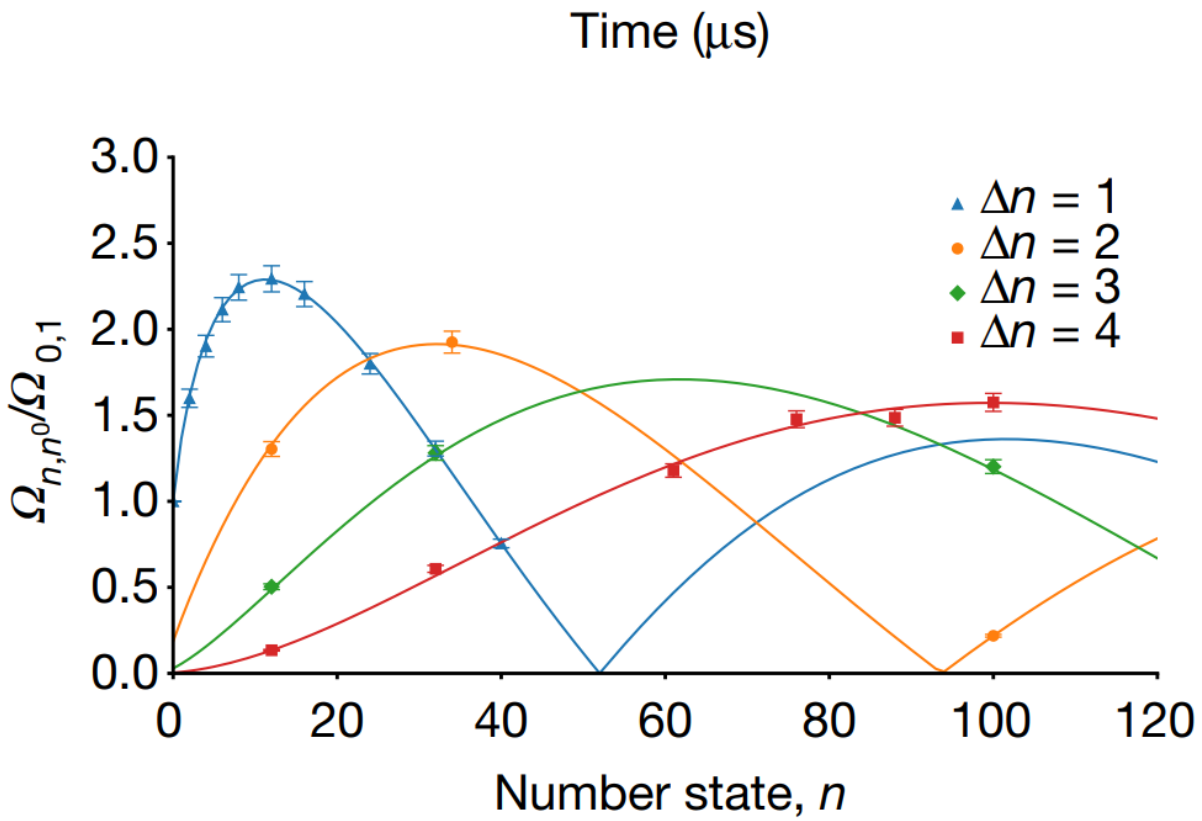


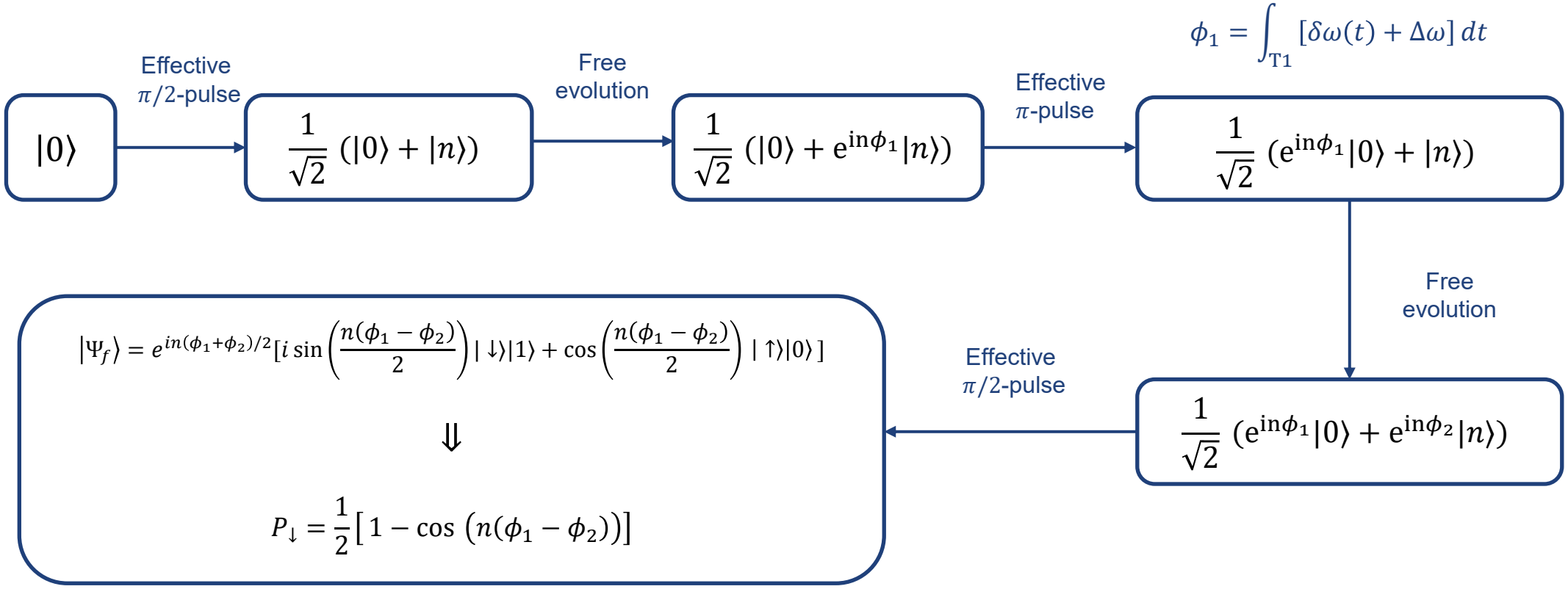
Fig. S.3: First- to fourth-order sideband Rabi frequencies. (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Rabi frequencies of higher order sidebands are given by

$$\frac{\Omega_{n,n+s}}{\Omega_{0,1}} = \eta^{|s|-1}\sqrt{\frac{n_<!}{n_>!}}L_{n_<}^{|s|}(\eta^2)$$

- η is the Lamb-Dicke parameter
- $L_n^{|s|}$ is the generalized Laguerre polynomial
- $n_>(n_<)$ is the greater (smaller) of $n+s$ and n

Details of the echo pulse



$$n(\phi_1 - \phi_2) = n\int_1 \delta\omega\, dt - n\int_2 \delta\omega\, dt + n\,\Delta\omega\,(T_1 + T_2)$$

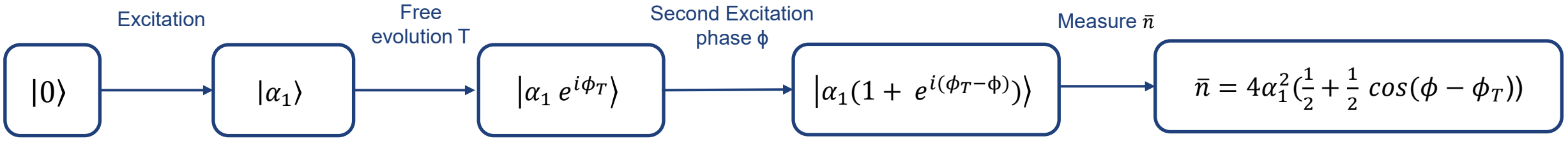
Fixing resources – Classical vs Quantum comparison

The classical limit to measurement precision of the frequency of a harmonic oscillator is $\delta\phi \propto 1/\sqrt{\bar{n}}$, where $\bar{n}$ is the average occupation number. But what is the prefactor?

- According to R.J. Glauber, a classical protocol uses only coherent states and displacements

  $\Rightarrow$ Shot noise limited $\Delta P = \sqrt{\bar{n}}$

- To compare to number state interferometers, require that $\bar{n} \leq n$, that is the classical interferometer does not use more energy than the quantum one



For fair comparison, we restrict $4\alpha_1^2 = \mathrm{n}$, and for a small $\phi_T \simeq 0$, we have maximal signal at $\phi = \pi/2$ and

$$\delta\phi = \frac{\Delta P}{|\partial P_0/\partial\phi|} = \sqrt{\frac{2}{n}}$$

Auto-balanced Ramsey protocol

Raman pulses are used for sidebands ⇒ Stark shift adds a systematic phase offset

$$\phi_{\text{sys}} \approx \int_0^{t_{pulse}} \frac{\Delta E_{Stark}}{\hbar}\, dt \text{ + initial phase}$$

4 Ramsey experiments used, each pair with:

$$T_{short} = 20\mu s \Rightarrow \text{Phase offset}$$
$$T_{long} = 100\mu s \Rightarrow \text{Frequency offset}$$

The total acquired phase will give an error signal:

$$\Delta P \approx -n\delta\omega T + \phi_{sys} + \delta\phi_{LO}$$

$$T_{short} \Rightarrow \delta\phi_{LO} = n\delta\omega T_{short} - \phi_{sys}$$
$$T_{\text{long}} \Rightarrow \Delta P = -n\delta\omega\left(T_{long} - T_{short}\right)$$

→ Adjust the frequency ω_{LO}

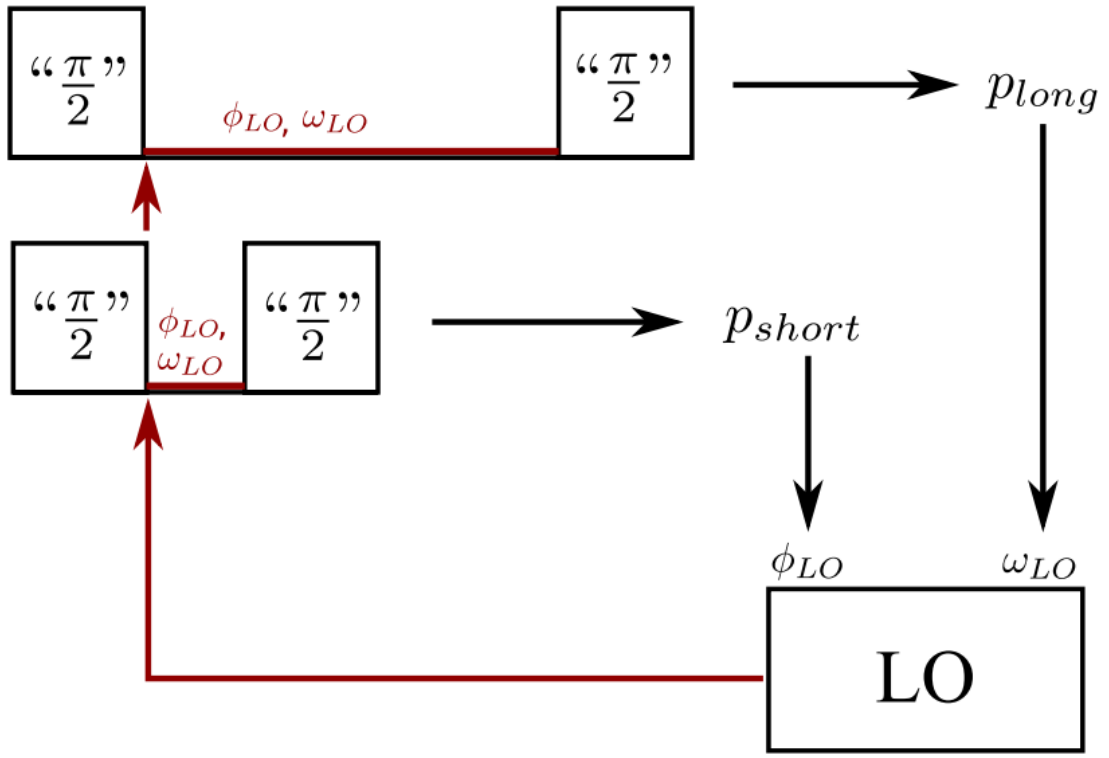


Figure S.4: Schematic illustrating the auto-balanced feedback loop (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Allan deviation

- Standard deviation fails for a drifting signal
- **Allan deviation** measures stability vs averaging time τ

- Split frequency record into consecutive bins of length τ, average within each one to get a sequence of values $\bar{y}_i(\tau)$, and then take the variance of the differences between adjacent bins:

$$\sigma(\tau) = \sqrt{\frac{1}{2}\langle(\overline{y_{i+1}} - \bar{y}_i)^2\rangle}$$

  - Minimum: optimal averaging time

Tracking speed: feedback vs no feedback (n=8)

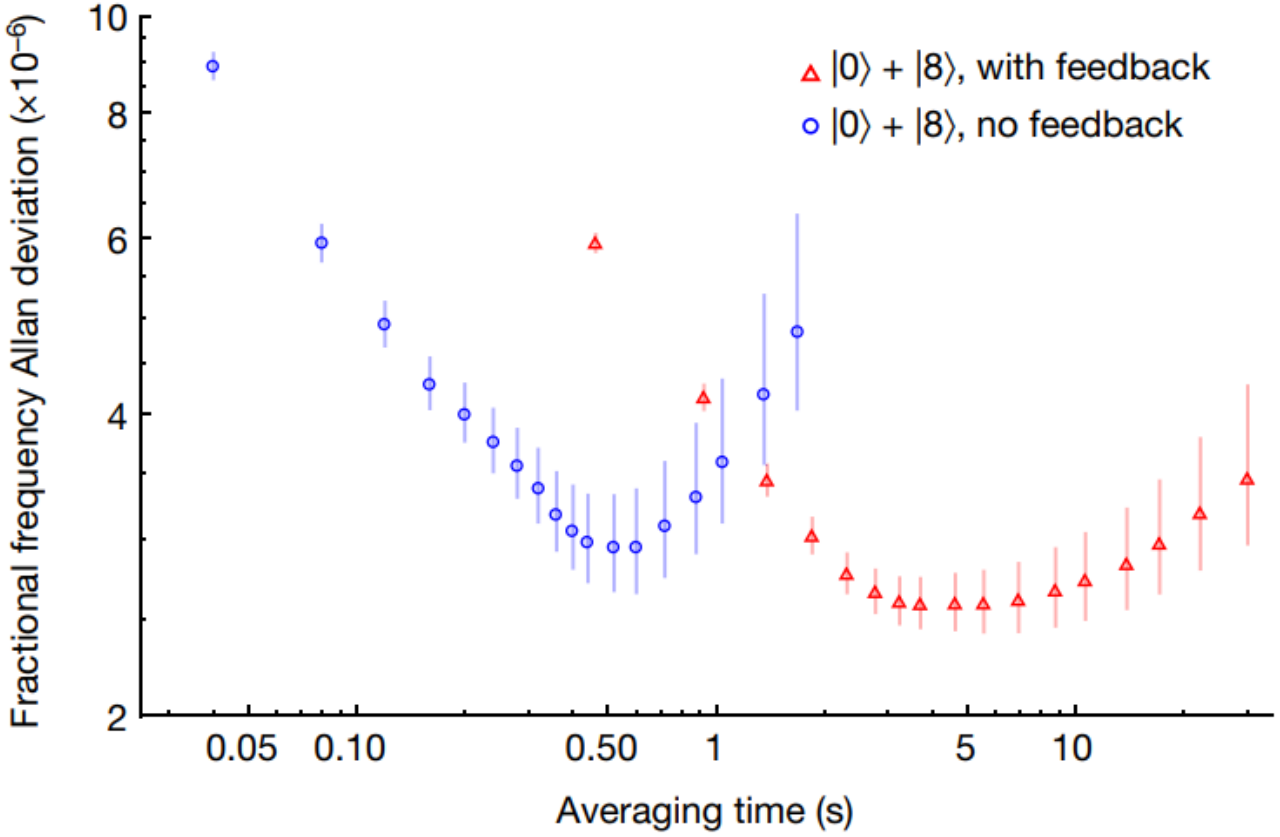


Figure S.5: Uncertainty of the fractional mode frequency versus averaging time for two series of only n = 8 interferometer runs performed to maximize the measurement duty cycle. The Allan standard deviation for averaging times of up to 1 s, without tracking activated, is shown by the blue circles.(Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))

Sources of decoherence

- Heating rate is approximately low, 17 quanta per second

- Dominant source of decoherence is dephasing due to fluctuations in the mode frequency
  - Technical noise from voltage source
  - Uncontrolled charging from stray light scattering off dielectric material between trap electrodes
  - Amplitude instabilities in rf source

- Longer timescale (minutes,hours) drift attributed to uncontrolled charging and discharging, that affects frequency tracking

- Detailed investigation is currently in progress

Experimental Timescales

State preparation:

1. Doppler cooling: 120μs
2. Sideband cooling: 110μs
3. Microwave carrier π-pulse : 5μs
4. Sideband π-pulses, depending on n

Readout: 400μs detection pulse

n	$\mathrm{T}(\lvert n\rangle)$ (μs)	$\mathrm{T}(\lvert 0\rangle + \lvert n\rangle)$ (μs)
2	22	16
4	38	41
6	51	54
8	63	66
12	86	89
16	109	112
20	133	–
40	196	–
80	293	–
100	335	–

Table S.1: Duration of pulse sequences used to produce number states and number-state superpositions (Extracted from McCormick et al., "Quantum-enhanced sensing of a single-ion mechanical oscillator," Nature 572, 86–90 (2019))